

Problem 1.

X legitimate and Y spam email.

$$X \sim \text{Pois}(\lambda_1) \quad , \quad Y \sim \text{Pois}(\lambda_2)$$

$$T = X + Y$$

$$P(T=n) = P(X+Y=n) = \sum_{k=0}^n P(X=k) P(Y=n-k)$$

$$= \sum_{k=0}^n e^{-\lambda_1} \frac{\lambda_1^k}{k!} e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!}$$

$$= e^{-\lambda_1 - \lambda_2} \sum_{k=0}^n \frac{\lambda_1^k \lambda_2^{n-k}}{k! (n-k)!}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} \sum_{k=0}^n \frac{n!}{k! (n-k)!} \lambda_1^k \lambda_2^{n-k}$$

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} (\lambda_1 + \lambda_2)^n$$

$$\Rightarrow T \sim \text{Pois}(\lambda_1 + \lambda_2)$$

$$P(X=k | T=n) = \frac{P(X=k, Y=n-k)}{P(T=n)} = \frac{e^{-\lambda_1} \frac{\lambda_1^k}{k!} \frac{\lambda_2^{n-k}}{(n-k)!} e^{-\lambda_2}}{e^{-\lambda_1 - \lambda_2} \frac{(\lambda_1 + \lambda_2)^n}{n!}}$$

$$= \frac{n!}{k! (n-k)!} \frac{\lambda_1^k \lambda_2^{n-k}}{(\lambda_1 + \lambda_2)^n}$$
$$P(X=k | T=n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

$$X | T=n \sim \text{Bin}(n, \frac{\lambda_1}{\lambda_1 + \lambda_2})$$

$$E[X | T=n] = n \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

Problem 2. See discussion 6, question 5.

Problem 3. See discussion 6, question 2

problem 4. See discussion 6, question 1

problem 5

X and Y are independent

$$X \sim \text{uniform}(0,1)$$

$$Y \sim \text{uniform}(0,1)$$

$$\begin{aligned} F_Z(z) &= P(Z \leq z) = P(\max\{2X, Y\} \leq z) \\ &= P(2X \leq z, Y \leq z) \end{aligned}$$

X and Y are independent

$$\begin{aligned} &= P(2X \leq z) P(Y \leq z) \\ &= P(X \leq \frac{z}{2}) P(Y \leq z) \\ &= F_X(\frac{z}{2}) F_Y(z) \end{aligned}$$

$$F_Z(z) = \frac{1}{2} F_X(\frac{z}{2}) F_Y(z) + F_X(\frac{z}{2}) F_Y(z)$$

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ y & 0 < y < 1 \\ 1 & y > 1 \end{cases}$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 < x < 1 \\ 1 & x > 1 \end{cases}$$

if $0 < z < 1$

$$f_z(z) = \frac{1}{2} \cdot 1 \cdot z + \frac{z}{2} \cdot 1 = z$$

if $1 < z < 2$

$$f_z(z) = \frac{1}{2} \cdot 1 \cdot 1 + \frac{z}{2} \cdot 0 = \frac{1}{2}$$

$$f_z(z) = \begin{cases} z & 0 < z < 1 \\ \frac{1}{2} & 1 < z < 2 \\ 0 & \text{otherwise} \end{cases}$$

problem 6

$$X \sim N(0, 1)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$x \in \mathbb{R}.$$

$$g(t) = \begin{cases} -t & t \leq 0 \\ \sqrt{t} & t > 0 \end{cases}$$

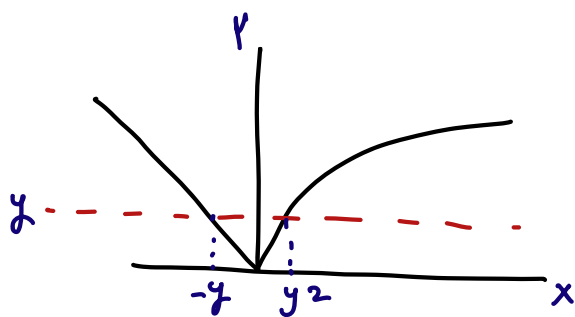
The range of $g(t)$ is always positive.

$$Y = g(X)$$

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) \quad \text{if } y > 0$$

$$= P(-y \leq X \leq y^2) \quad \text{if } y \geq 0$$

$$= F_X(y^2) - F_X(-y) \quad \text{if } y \geq 0$$



$$\Rightarrow f_Y(y) = f_X(y^2) (2y) + f_X(-y) \quad \text{if } y > 0$$

$$\Rightarrow f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^4}{2}} 2y + \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \quad \text{if } y > 0$$

$$f_Y(y) = \begin{cases} \frac{1}{\sqrt{2\pi}} \left(2y e^{-\frac{y^4}{2}} + e^{-\frac{y^2}{2}} \right) & y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Problem 7.

$$Z = X - Y, \quad \text{where}$$

$$X = Y = \text{Dis uniform } (\{1, 2, 3, \dots, 6\})$$

$$\begin{aligned} E[Z^2] &= E[(X-Y)^2] = E[X^2 - 2XY + Y^2] \\ &= E[X^2] - 2E[XY] + E[Y^2] \end{aligned}$$

$$\text{since } X \text{ and } Y \text{ are independent } E[XY] = E[X]E[Y]$$

$$E[Z^2] = E[X^2] - 2E[X]E[Y] + E[Y^2]$$

since X and Y have same distribution

$$E[X] = E[Y]$$

$$E[X^2] = E[Y^2]$$

$$\begin{aligned} E[Z^2] &= 2E[X^2] - 2(E[X])^2 = 2(E[X^2] - (E[X])^2) \\ &= 2 \text{var}(X) \underset{\uparrow}{=} 2 \frac{(6-1+1)^2 - 1}{12} = \frac{35}{6}. \end{aligned}$$

$X \sim \text{Disuniform}$

using formula

Problem 8.

$$X_i \sim \text{Exp}(\lambda_i)$$

X_i are i.i.d

$$Y = \min(X_1, X_2, \dots, X_n)$$

we will use cdf method.

$$F_Y(y) = P(Y \leq y) = 1 - P(Y > y)$$

$$= 1 - P(\min(X_1, X_2, \dots, X_n) > y)$$

$$= 1 - P(X_1 > y, X_2 > y, \dots, X_n > y)$$

$$\stackrel{\uparrow}{=} 1 - P(X_1 > y) P(X_2 > y) \dots P(X_n > y)$$

independence

For a exponential random variable $X \sim \text{Exp}(\lambda)$

$$\begin{aligned} P(X > a) &= \int_a^\infty \lambda e^{-\lambda t} dt = -e^{-\lambda t} \Big|_a^\infty \\ &= e^{-a\lambda} \end{aligned}$$

$$\text{Hence, } F_Y(y) = 1 - e^{-\lambda_1 y} \cdot e^{-\lambda_2 y} \dots e^{-\lambda_n y}$$

$$F_Y(y) = 1 - e^{-(\lambda_1 + \lambda_2 + \dots + \lambda_n)y}$$

Now we take derivative with respect to y from both side,

$$f_Y(y) = (\lambda_1 + \lambda_2 + \dots + \lambda_n) e^{-(\lambda_1 + \lambda_2 + \dots + \lambda_n)y}$$

$$\Rightarrow Y \sim \text{EXP}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$$

Problem 9.

20 coders

10 startup companies

10 breakout room with 2 spaces each.

$$X = I_1 + I_2 + \dots + I_{10}$$

$$\text{var}(X) = \text{Cov}\left(\sum_{i=1}^{10} I_i, \sum_{j=1}^{10} I_j\right) = \sum_{i=1}^{10} \sum_{j=1}^{10} \text{Cov}(I_i, I_j)$$

$$\text{var}(X) = \sum_{i=1}^{10} \text{var}(I_i) + 2 \sum_{i < j} \text{Cov}(I_i, I_j) \quad (*)$$

$$\text{var}(I_i) = p(1-p) \quad \text{where} \quad p = P(I_i=1)$$

$$\text{since } I_i \sim \text{Bern}(p)$$

$$p = P(I_i=1) = \frac{20}{20} \cdot \frac{1}{19} = \frac{1}{19}$$

$$\text{var}(I_i) = \frac{18}{(19)^2}$$

$$I_i I_j = \begin{cases} 1 & \text{if } I_i=1 \text{ and } I_j=1 \\ 0 & \text{otherwise} \end{cases}$$

Equation $(*)$ can be simplified further to

$$\text{var}(X) = 10 \text{var}(I_1) + 2 \binom{10}{2} \text{Cov}(I_1, I_2)$$

$$\text{var}(X) = \frac{180}{(19)^2} + 2 \frac{10 \cdot 9}{2} \text{Cov}(I_1, I_2) \quad **$$

$$I_1, I_2 \sim \text{Bern}(P(I_1=1 \text{ and } I_2=1))$$

$$P(I_1=1 \text{ and } I_2=1) = P(I_2=1 | I_1=1) P(I_1=1)$$

$$\sim = \frac{18}{18} \cdot \frac{1}{17} \cdot \frac{1}{19} = \frac{1}{17} \cdot \frac{1}{19}$$

$$E[I_1, I_2] = \frac{1}{17} \cdot \frac{1}{19}, \quad E[I_1] E[I_2] = \left(\frac{1}{19}\right)^2$$

$$\text{Cov}(I_1, I_2) = E[I_1, I_2] - E[I_1] E[I_2]$$

$$\Rightarrow \text{Cov}(I_1, I_2) = \frac{2}{(19)^2 \cdot 17}$$

Hence, from ***

$$\text{Var}(X) = \frac{180}{(19)^2} + 10 \cdot 9 \cdot \frac{2}{(19)^2 \cdot 17}$$

$$\text{Var}(X) = \frac{180}{(19)^2} \left[1 + \frac{1}{17} \right] = \left(\frac{18}{19} \right)^2 \frac{10}{17}$$

problem 10.

$$P(X=3 \text{ or } X=4) = P(X=3) + P(X=4)$$

$$= \frac{365 \cdot 364 \cdot \binom{2}{1}}{(365)^3} + \frac{365 \cdot 364 \cdot 363 \cdot \binom{3}{1}}{(365)^4}$$

$$= \frac{364}{365} \cdot \frac{2}{365} + \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{3}{365}$$

Problem 11.

$A = 6$ Heads in first 8 tosses.

$B =$ The 9th trial results in heads

- (a) trials are independent from each other
hence what happens in the 9th trial
is independent of what has happened in the
first 8 trials, hence

$$P(B|A) = P(B) = p$$

- (b) $E = 3$ heads in the first 4 trials
 $F = 2$ heads in the last 3 trials

E and F are independent

$$P(E \cap F) = P(E) \cdot P(F)$$

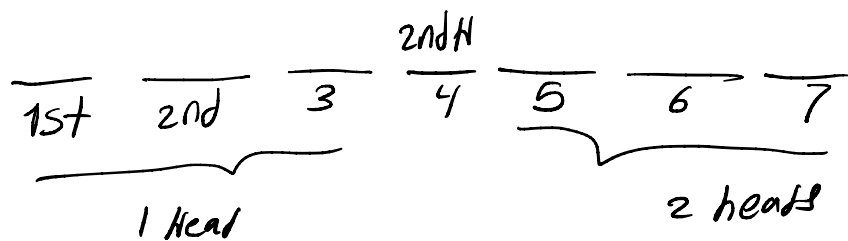
$$P(E \cap F) = \binom{4}{3} p^3 (1-p) \binom{3}{2} p^2 (1-p)$$

$$P(E \cap F) = \binom{4}{3} \binom{3}{2} p^5 (1-p)^2$$

(c) $A = 4$ Heads in first 7 tosses

$B = 2$ nd head occurs at 4th toss

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{\binom{3}{1} \binom{3}{2} p^4 (1-p)^3}{\binom{7}{4} p^4 (1-p)^3}$$



$$P(B|A) = \frac{3 \cdot 3}{35} = \frac{9}{35}$$

(d) $H_1 = 5$ Heads in the first 6 tosses

$H_2 = 3$ Heads in the last 5 tosses

$E =$ the 6th toss is Head

$E^c =$ the 6th toss is not head.

$$P(H_1 \cap H_2) = P(H_1 \cap H_2 | E) P(E) + P(H_1 \cap H_2 | E^c) P(E^c)$$

$$P(H_1 \cap H_2) = \binom{5}{4} p^4 \cdot p \binom{4}{2} p^2 (1-p)^3 + \binom{5}{5} p^5 (1-p) \binom{4}{3} p^3 (1-p)$$

$$\underbrace{\qquad\qquad\qquad}_{\substack{\text{6th} \\ \text{Head}}} \qquad\qquad\qquad \underbrace{\qquad\qquad\qquad}_{\substack{\text{6th} \\ \text{T}}}$$

$$P(H_1 \cap H_2) = 30 p^7 (1-p)^3 + 4 p^8 (1-p)^2$$

Problem 12.

Define
$$I_i = \begin{cases} 1 & \text{the } i\text{th person picks their own name} \\ 0 & \text{otherwise} \end{cases}$$

$$I_i \sim \text{Bern}(p = P(I_i = 1))$$

$$P(I_i = 1) = \frac{1}{n} \quad \Rightarrow \quad \text{var}(I_i) = p \cdot (1-p) = \frac{1}{n} \frac{(n-1)}{n}$$

$$\text{Cov}(I_i, I_j) = E[I_i I_j] - E[I_i] E[I_j]$$

$$I_i I_j = \begin{cases} 1 & \text{if } I_i = 1 \text{ \& } I_j = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$I_i I_j \sim \text{Bern}(p_2 = P(I_i = 1 \text{ and } I_j = 1))$$

$$\begin{aligned} p_2 = P(I_i = 1 \text{ and } I_j = 1) &= P(I_j = 1 \mid I_i = 1) P(I_i = 1) \\ &= \frac{1}{n-1} \cdot \frac{1}{n} \end{aligned}$$

$$\begin{aligned} \text{Cov}(I_i, I_j) &= \frac{1}{n-1} \cdot \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n} \left[\frac{1}{n-1} - \frac{1}{n} \right] \\ &= \frac{1}{n^2(n-1)} \end{aligned}$$

X := number of people who pick their own name.

$$X = \sum_{i=1}^n I_i$$

$$E[X] \underset{\substack{\nearrow \\ \text{linearity}}}{=} \sum_{i=1}^n E[I_i] = n \cdot \frac{1}{n} = 1$$

$$\text{Var}(X) = \text{Cov}\left(\sum_{i=1}^n I_i, \sum_{j=1}^n I_j\right) = \sum_{i=1}^n \text{Var}(I_i) + 2 \binom{n}{2} \text{Cov}(I_1, I_2)$$

$$\begin{aligned} \text{Var}(X) &= \frac{n-1}{n^2} \cdot n + 2 \frac{n(n-1)}{2} \frac{1}{n^2(n-1)} \\ &= \frac{n-1}{n} + \frac{1}{n} = \frac{n-1+1}{n} = 1 \end{aligned}$$